

Peiyao Sheng

peiyaosheng@gmail.com | [simplespy.github.io](https://github.com/simplespy) | [Google Scholar](#)

Education

University of Illinois Urbana-Champaign

M.S. (2021) and Ph.D. (2024) in Computer Science

2019 – 2024

Advisor: Prof. Pramod Viswanath

Princeton University

Visiting Researcher

Jul 2022 – Nov 2024

Host: Prof. Pramod Viswanath

Shanghai Jiao Tong University

B.S. in Computer Science and Technology

2015 – 2019

ACM Honors Class (elite CS program for the top 5% of students), Zhiyuan College

Research Interests

Trustworthiness of LLMs and agents: verifiable reasoning, LLM evaluation, agent security, and model fingerprinting. Background in the security and accountability of decentralized systems. accountability of decentralized systems.

Work Experience

Industry

Sentient Labs

Dec 2024 – Present

Researcher

- Verifiable reasoning: built tool-augmented evaluation of LLM reasoning, process supervision that rewards sound reasoning, and representation-aligned supervision toward faithful world models on chess testbed.
- Agent security: designed memory-injection attacks on deployed Web3 agents, and cross-user poisoning attacks on multi-user collaborative agents.
- Contributed to scalable LLM fingerprinting with a public [leaderboard](#), and to LiveCodeBench Pro, an olympiad-level coding benchmark.
- Organized Open AGI Symposiums at NeurIPS 2025 and UC Berkeley.

Witness Chain

2022 – 2024

Co-founder & Head of Research

- Established the research foundation for a verifiable physical trust stack: proof of location, proof of backhaul and proof of diligence.
- Built a unified coordination layer for decentralized physical infrastructure networks (DePINs), enabling trust-less coordination and shared value creation.
- Deployed across 6,000+ nodes worldwide and secured by over \$5B in restaked assets; the company raised an \$8M seed round.

Chainlink Labs

Summer 2023

Research Intern

Mentor: Dr. Dahlia Malkhi

- Developed “ticketing” for orchestrating atomic broadcasts across nodes, including a hybrid of managed and unmanaged regimes balancing adaptivity and resilience.

Kaleidoscope Blockchain

Summer 2021

Research Intern

- Built a prototype of a tokenized telco marketplace connecting network applications with infrastructure, with on-chain enforcement of payments and service-level agreements.

Publications

[†] Co-mentored project

AI & Language Models

- [1] *Cross-User Poisoning: User-Task Boundary Failures in Multi-User Collaborative Language Agents*
Atharv Singh Patlan, **Peiyao Sheng**[†], S. Ashwin Hebbar, Prateek Mittal, Pramod Viswanath
Conference on Neural Information Processing Systems (NeurIPS), 2026
- [2] *Correct Answers from Sound Reasoning: Verifiable Process Supervision for Language Models*
Kyuyoung Kim, Kevin Wang, Yunfei Xie, Peiyang Xu, **Peiyao Sheng**, Chen Wei, Zhangyang Wang, Jinwoo Shin, Pramod Viswanath, Sewoong Oh
Conference on Language Modeling (COLM), 2026
- [3] *Representation-Aligned Auxiliary Supervision for Language Model Adaptation*
Kyuyoung Kim, **Peiyao Sheng**[†], Ashwin Hebbar, Peiyang Xu, Yunfei Xie, Kevin Wang, Rui Xin, Chen Wei, Zhangyang Wang, Jinwoo Shin, Pramod Viswanath, Sewoong Oh
NeurIPS 2026 Workshop on Linguistic Principles for Foundation Models
Outstanding Runner-Up Award
- [4] *Hallucinations on the Board: Tool-Augmented Evaluation of LLM Chess Commentary*
Ashwin Hebbar, **Peiyao Sheng**[†], Sewoong Oh, Pramod Viswanath
Preprint, 2026
- [5] *Real AI Agents with Fake Memories: Fatal Context Manipulation Attacks on Web3 Agents*
Atharv Singh Patlan, **Peiyao Sheng**[†], S. Ashwin Hebbar, Prateek Mittal, Pramod Viswanath
Financial Cryptography and Data Security (FC), 2026
- [6] *LiveCodeBench Pro: How Do Olympiad Medalists Judge LLMs in Competitive Programming?*
Zihan Zheng, Zerui Cheng, Zeyu Shen, Shang Zhou, Kaiyuan Liu, Hansen He, Dongruixuan Li, Stanley Wei, Hangyi Hao, Jianzhu Yao, **Peiyao Sheng**, Zixuan Wang, Wenhao Chai, Aleksandra Korolova, Peter Henderson, Sanjeev Arora, Pramod Viswanath, Jingbo Shang, Saining Xie
Conference on Neural Information Processing Systems (NeurIPS), 2025
- [7] *Scalable Fingerprinting of Large Language Models*
Anshul Nasery, Jonathan Hayase, Creston Brooks, **Peiyao Sheng**, Himanshu Tyagi, Pramod Viswanath, Sewoong Oh
Conference on Neural Information Processing Systems (NeurIPS), 2025
- [8] *CHANCERY: Evaluating Corporate Governance Reasoning Capabilities in Language Models*
Lucas Irwin, Arda Kaz, **Peiyao Sheng**[†], Pramod Viswanath
Preprint, 2025
- [9] *GANs for Children: A Generative Data Augmentation Strategy for Children Speech Recognition*
Peiyao Sheng, Zhuolin Yang, Yanmin Qian
IEEE Automatic Speech Recognition and Understanding Workshop (ASRU), 2019
- [10] *Data Augmentation using Conditional Generative Adversarial Networks for Robust Speech Recognition*
Peiyao Sheng, Zhuolin Yang, Hu Hu, Tian Tan, Yanmin Qian
International Symposium on Chinese Spoken Language Processing (ISCSLP), 2018

Decentralized Systems & Security

- [11] *Unconditionally Safe Light Client*
Niusha Moshrefi, **Peiyao Sheng**[†], Soubhik Deb, Sreeram Kannan, Pramod Viswanath
International Conference on Blockchain (ICBC), 2025
Best Paper Award
- [12] *On Orchestrating Parallel Broadcasts for Distributed Ledgers*
Peiyao Sheng, Chenyuan Wu, Dahlia Malkhi, Michael K. Reiter, Chrysoula Stathakopoulou, Michael Wei, Maofan Yin
Preprint, 2024

- [13] *Proof of Diligence: Cryptoeconomic Security for Rollups*
Peiyao Sheng, Ranvir Rana, Himanshu Tyagi, Pramod Viswanath
Advances in Financial Technologies (AFT), 2024
- [14] *CFT-Forensics: High-Performance Byzantine Accountability for Crash Fault Tolerant Protocols*
Weizhao Tang, **Peiyao Sheng**, Pronoy Roy, Xuechao Wang, Giulia Fanti, Pramod Viswanath
Advances in Financial Technologies (AFT), 2024
- [15] *BFT-PoLoc: A Byzantine Fortified Trigonometric Proof of Location Protocol using Internet Delays*
Peiyao Sheng, Vishal Sevani, Ranvir Rana, Himanshu Tyagi, Pramod Viswanath
Preprint, 2024
- [16] *Proof of Backhaul: Trustfree Measurement of Broadband Bandwidth*
Peiyao Sheng, Nikita Yadav, Vishal Sevani, Arun Babu, SVR Anand, Himanshu Tyagi, Pramod Viswanath
Network and Distributed System Security Symposium (NDSS), 2024
- [17] *SAKSHI: Decentralized AI Platforms*
Suma Bhat, Canhui Chen, Zerui Cheng, Zhixuan Fang, Ashwin Hebbar, Sreeram Kannan, Ranvir Rana, **Peiyao Sheng**, Himanshu Tyagi, Pramod Viswanath, Xuechao Wang
Preprint, 2023
- [18] *TrustBoost: Boosting Trust among Interoperable Blockchains*
Peiyao Sheng, Xuechao Wang, Kartik Nayak, Sreeram Kannan, Pramod Viswanath
ACM Conference on Computer and Communications Security (CCS), 2023
- [19] *Player-Replaceability and Forensic Support are Two Sides of the Same (Crypto) Coin*
Peiyao Sheng, Gerui Wang, Kartik Nayak, Sreeram Kannan, Pramod Viswanath
Financial Cryptography and Data Security (FC), 2023
- [20] *Trust-free Service Measurement and Payments for Decentralized Cellular Networks*
S.V.R. Anand, Serhat Arslan, Rajat Chopra, Sachin Katti, Milind Kumar Vaddiraju, Ranvir Rana, **Peiyao Sheng**, Himanshu Tyagi, Pramod Viswanath (alphabetical order)
ACM Workshop on Hot Topics in Networks (HotNets), 2022
- [21] *BFT Protocol Forensics*
Peiyao Sheng, Gerui Wang, Kartik Nayak, Sreeram Kannan, Pramod Viswanath
ACM Conference on Computer and Communications Security (CCS), 2021
Best Paper Runner-Up
- [22] *ACeD: Scalable Data Availability Oracle*
Peiyao Sheng, Bowen Xue, Sreeram Kannan, Pramod Viswanath
Financial Cryptography and Data Security (FC), 2021

Honors & Awards

2026	Outstanding Runner-Up Award, NeurIPS 2026 Workshop on Linguistic Principles for Foundation Models
2025	Best Paper Award, International Conference on Blockchain (ICBC)
2022	Chainlink Labs Fellowship, Honorable Mention
2021	Best Paper Runner-Up, ACM Conference on Computer and Communications Security (CCS)
2020	ICPC Mid-Central USA Regional Contest, 2nd place
2019	Outstanding Graduate, Shanghai Jiao Tong University
2017	Kai Yuan Scholarship (top 2%)
2017	Meritorious Winner, MCM/ICM (top 10%)
2016, 2017	Zhiyuan Honorary Scholarship
2015, 2016	Academic Excellence Scholarship (1st place)